ServiceNow Certified Application Developer

SmartInternz

Project Title

Automated Network Request Management in ServiceNow

Platform	ServiceNow
Application Area	Service Catalog and Flow Designer
Primary Catalog Item	Network Request
Backend Table	Network Database Table
Automation	Flow Designer
Document Type	Project Implementation Report

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1. Project Overview

Automated Network Request Management in ServiceNow is a service-management solution designed to capture, process, approve, and track network-related requests through a structured self-service catalog. The implementation combines Service Catalog configuration, reusable Variable Sets, Catalog UI Policies, a dedicated backend table, a relationship for approval information, and Flow Designer automation.

1.2. Project Objective

The main objective is to replace a manual request-handling process with a consistent workflow that captures complete information at submission time and automatically performs the required backend actions.

- Provide a clear self-service Network Request form for end users.

- Capture request-specific information through ten catalog variables.
- Reuse requester information through a dedicated Variable Set.
- Show or hide fields dynamically using Catalog UI Policies.
- Store submitted information in a Network Database Table.
- Associate request records with approval information.
- Automate record creation, approval, notification, and update activities using Flow Designer.
- Validate the implementation through flow testing and output verification.

1.3. End-to-End Process

1. User opens the ServiceNow catalog and selects Network Request.
2. Requester enters connection, relocation, location, device, and additional information.
3. Requester details are captured through the Requester information Variable Set.
4. Catalog UI Policies control fields according to the user's selections.
5. The submitted request is processed by the Service Catalog trigger.
6. Flow Designer retrieves catalog variables and creates a Network Database Table record.
7. Approval processing is initiated and the result is evaluated.
8. An email notification is sent as part of the approved processing path.
9. The backend record is updated and the execution is validated.

2. ServiceNow Start Building Page

The project begins in the ServiceNow Developer environment. The Start Building page provides access to application-development capabilities, while the ServiceNow home page and application navigator provide access to configuration areas used during implementation.

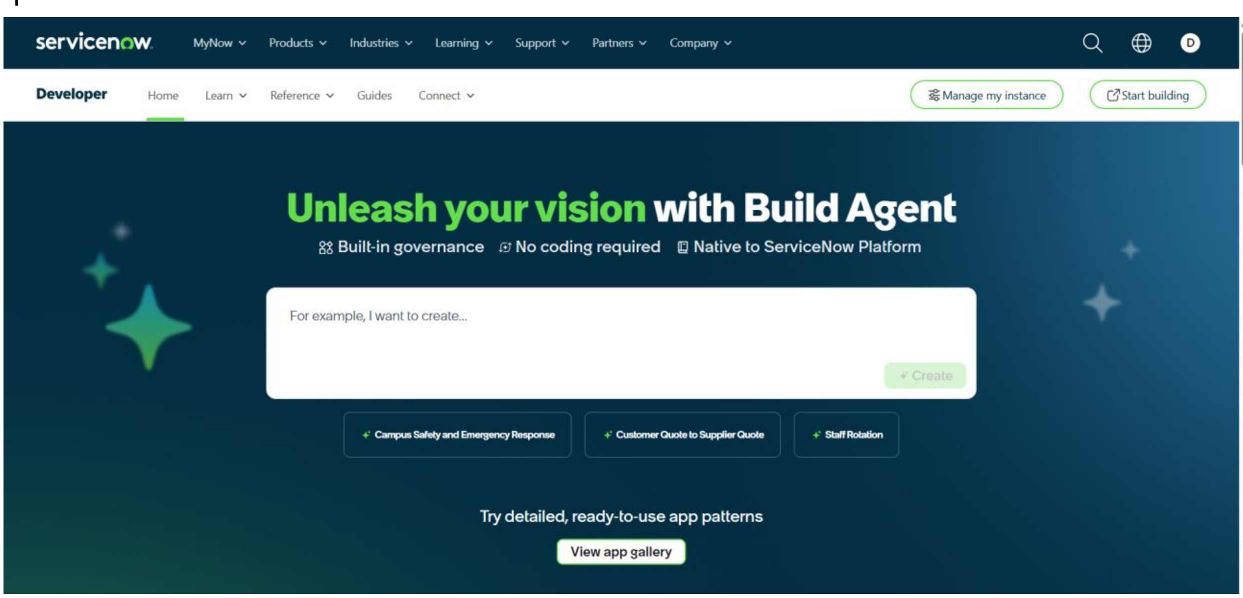


Fig. 1: ServiceNow Developer – Start Building page

3. ServiceNow Home / Application Navigation

- Open the ServiceNow Developer instance.
- Open the Start Building page.
- Go to the ServiceNow Home page.
- Click All in the Application Navigator.
- Use the filter/search box to find the required module.

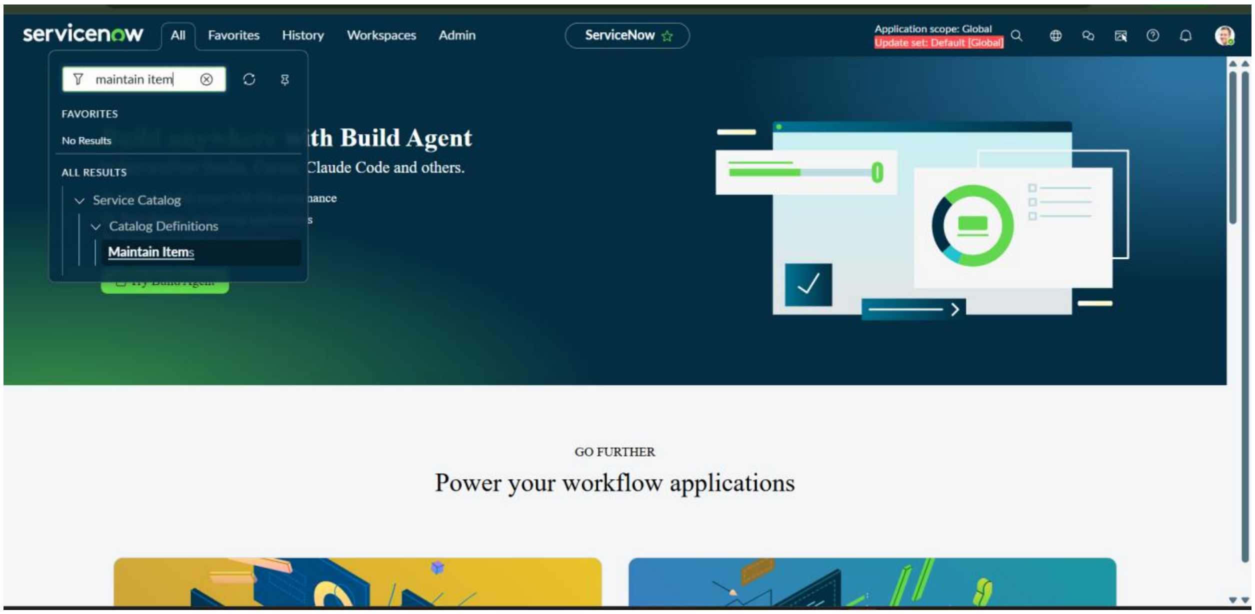


Fig. 2: ServiceNow home page and application navigation

Key Activities

- Open the ServiceNow Developer instance.
- From the Start Building page, open the required development/configuration area.
- Click All and search for the required module in the Application Navigator.
- Navigate to All > Service Catalog > Catalog Definitions > Maintain Items.

4. Creation of Catalog Item

Step 3: Navigate to All > Service Catalog > Catalog Definitions > Maintain Items. Search for Network Request and open the catalog item. This is the parent record for Variables, Variable Sets, and Catalog UI Policies.

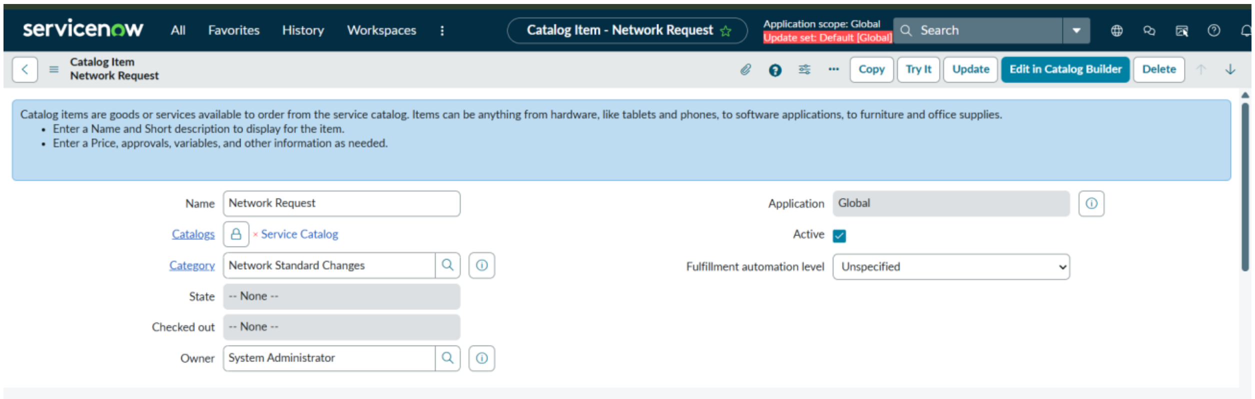


Fig. 3: Network Request Catalog Item configuration

Configuration Steps

10. Navigate to All > Service Catalog > Catalog Definitions > Maintain Items.
11. Search for Network Request and open it. If it does not exist, click New and create it.
12. Enter the required catalog item properties such as Name, Catalog, Category, Description, and other project-required values.
13. Click Submit/Update to save the catalog item.
14. Scroll to the related lists and configure Variables, Variable Sets, and Catalog UI Policies.

Role in the Solution

The catalog item is the entry point for the entire automation. Values submitted through this item become the input data for Flow Designer and are ultimately stored in the Network Database Table.

4.2. Variables Configuration – 10 Variables

- Click All.
- Go to Service Catalog → Catalog Definitions → Maintain Items.
- Open Network Request.
- Scroll to the Variables related list.
- Click New.
- Select the required Variable Type.
- Enter the Order number.
- Enter the Question / Label.
- Enter the variable Name.
- Configure any additional properties required.
- Click Submit.
- Repeat the same steps until all 10 variables are created.

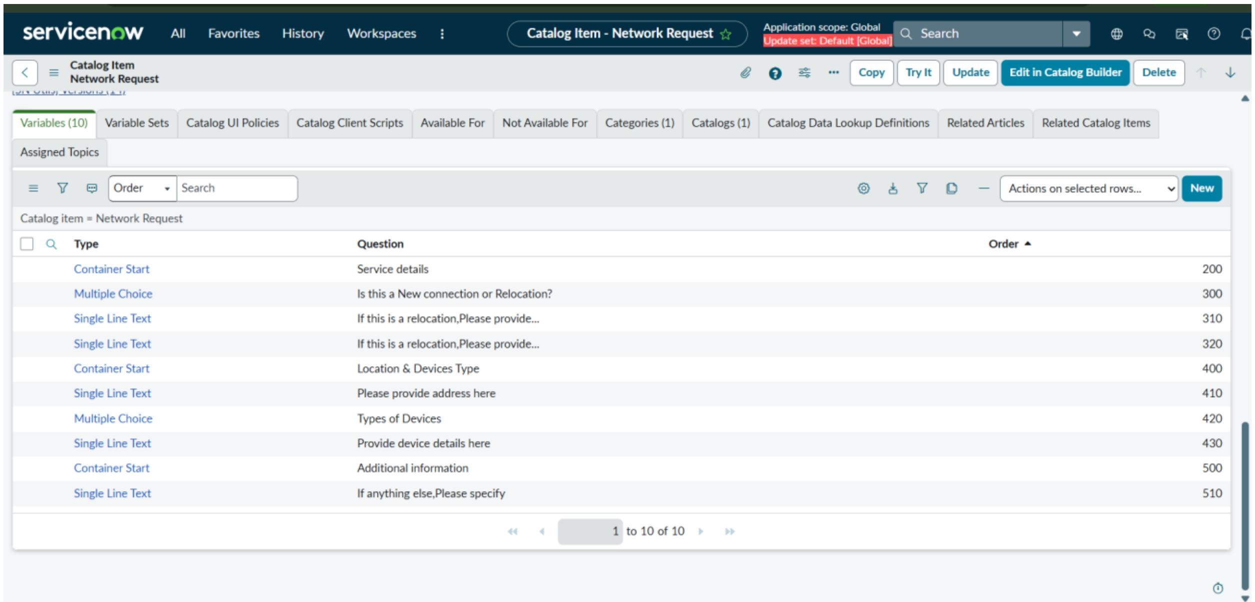


Fig. 4: Network Request – Variables related list (10)

Variable Configuration Principles

- Select the required variable Type, such as Single Line Text, Multiple Choice, Reference, or Attachment.
- Enter the Order value so the variables appear in the required sequence.
- Enter the Question/Label displayed to the requester.
- Enter a meaningful Name for later Flow Designer mapping.
- Use container variables where required to group related fields.

Project Variable Groups

Variable group	Purpose
Connection type	Captures whether the request is a New connection or Relocation.
Relocation details	Captures information required for a relocation request.
Location / address	Captures the address associated with the request.
Device information	Captures device type and device-specific details.
Additional information	Captures any remaining information required to process the request.

The screenshot shows the ten-variable configuration attached to the Network Request catalog item. These variables are processed before the reusable requester Variable Set is considered.

4.3. Variable Sets – Requester Information

Step 5: Open All > Service Catalog > Catalog Definitions > Maintain Items > Network Request. Scroll to Variable Sets and click New. Create Requester information, set its order, and click Submit. Open it and configure its variables.

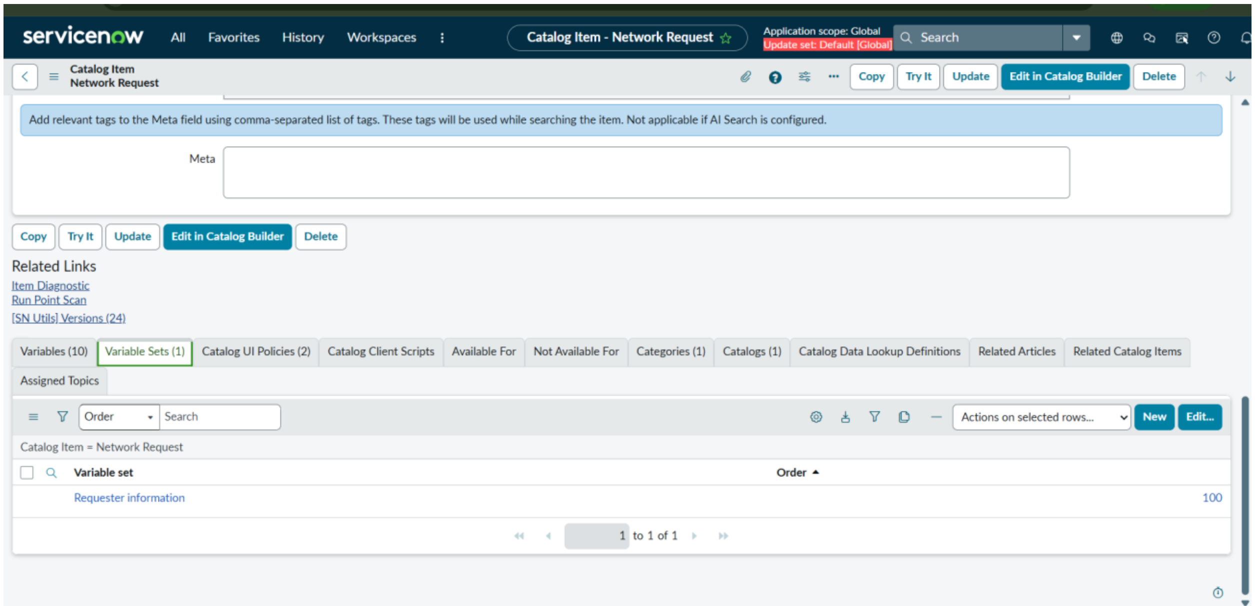


Fig. 5: Network Request – Variable Sets (1)

Requester Information Variable Set

The Requester information Variable Set contains five variables. Keeping these fields in a Variable Set makes the requester information reusable and logically separates it from network-request-specific data.

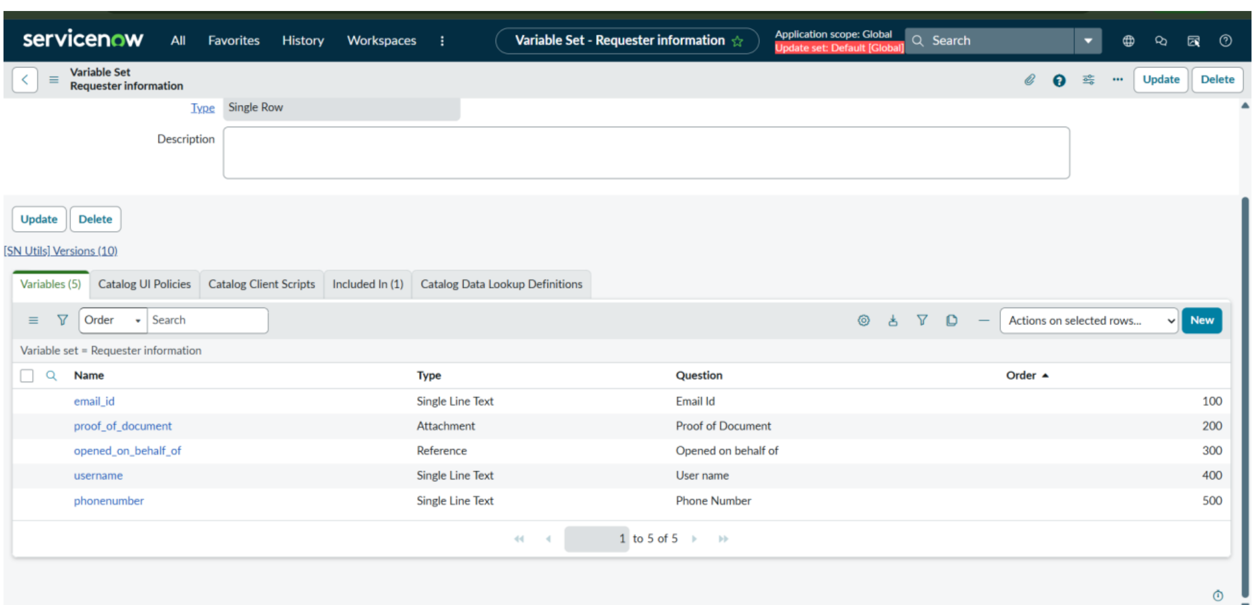


Fig. 6: Requester information Variable Set – Variables (5)

Variable	Purpose / type
email_id	Captures the requester's email address.

proof_of_document	Provides an attachment field for supporting documentation.
opened_on_behalf_of	References the user on whose behalf the request is opened.
username	Captures the requester's username.
phonenumber	Captures the requester's contact number.

Benefit

The Variable Set improves consistency because requester information is grouped in one reusable configuration instead of being mixed with the network-service fields.

4.4. Catalog UI Policies – 2 Policies

- Click All.
- Go to Service Catalog → Catalog Definitions → Maintain Items.
- Open Network Request.
- Scroll to Catalog UI Policies.
- Click New.
- Enter the UI Policy name.
- Select the required Catalog Item condition.
- Configure the UI Policy Actions.
- Click Submit / Update.
- Repeat the same process for the second Catalog UI Policy.
- Test the form to verify the fields show or hide correctly.

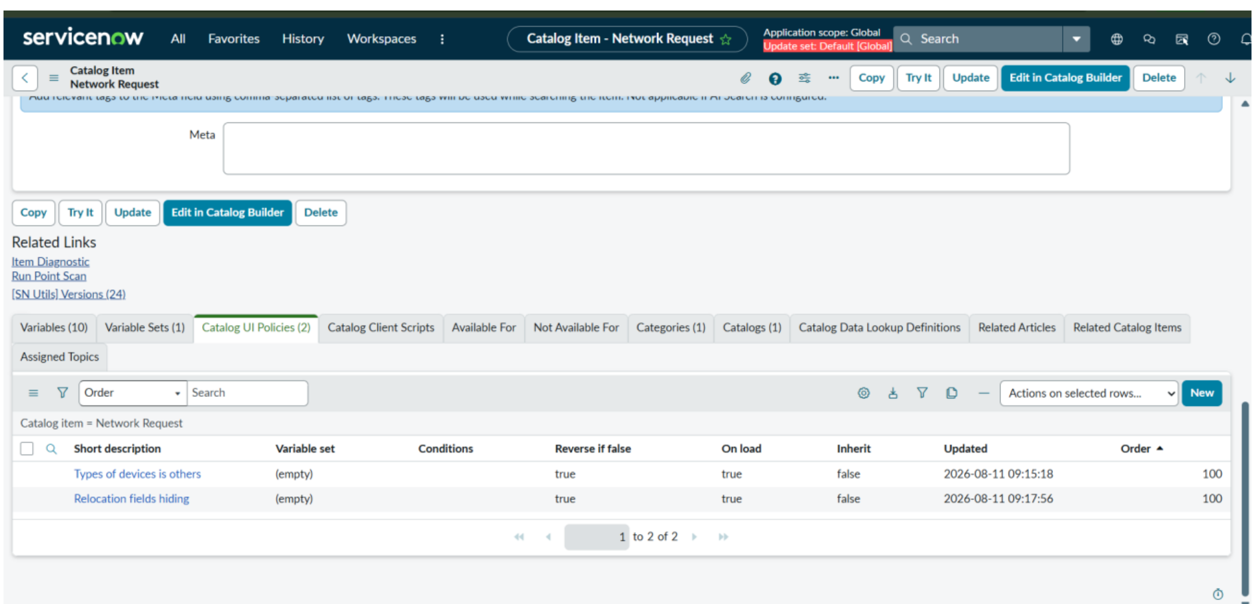


Fig. 7: Network Request – Catalog UI Policies (2)

Configured Policies

Policy	Purpose
Types of devices is others	Controls the behavior of additional device information when the requester selects Others.
Relocation fields hiding	Controls relocation-related fields based on the connection type selection.

Expected Form Behavior

- Fields that are not relevant to the selected request type should not unnecessarily distract the user.
- When Relocation is selected, the required relocation information can be made available.
- When the relevant device type is selected, the additional device field can be controlled accordingly.
- The policies improve usability and help prevent incomplete or irrelevant data entry.

5. Creation of Table

- Click All in the Application Navigator.
- Go to System Definition → Tables.
- Click New.
- In Label, enter Network Database Table.
- Verify the generated Name.
- Keep Application as Global if required by the project.
- Keep Create module selected if required.
- Enter the New menu name as Network Database Table if required.
- Review the table configuration.
- Click Submit.

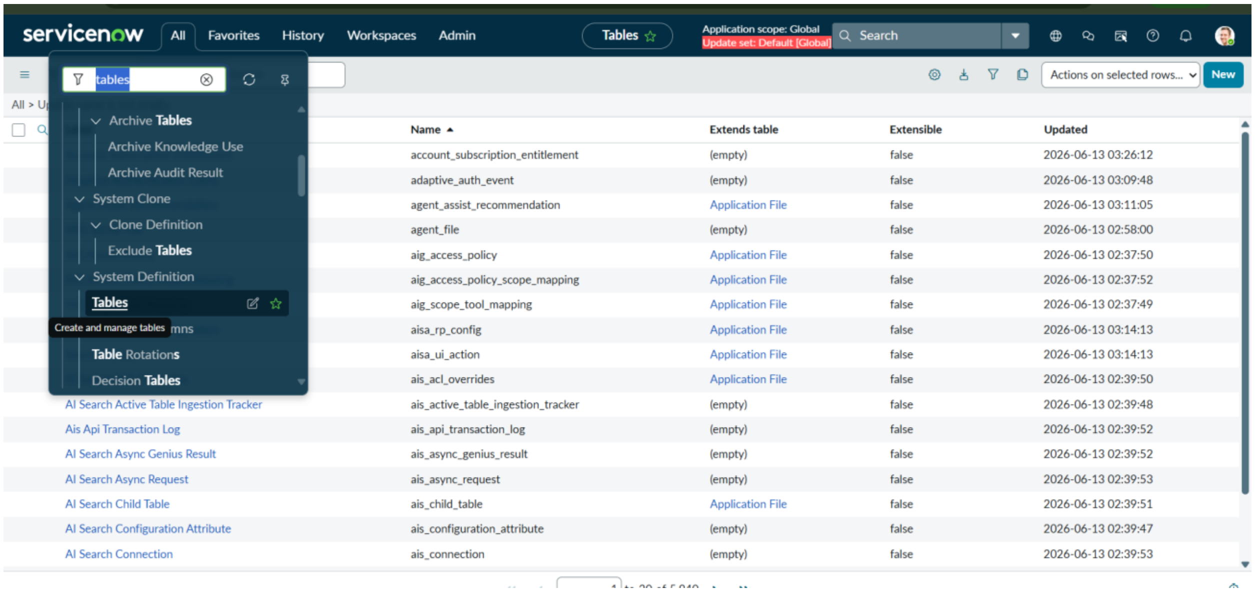


Fig. 8: ServiceNow Tables configuration / navigation

Create the Network Database Table

The table is created specifically for the project so that the network-request information can be stored independently and can be referenced by Flow Designer actions.

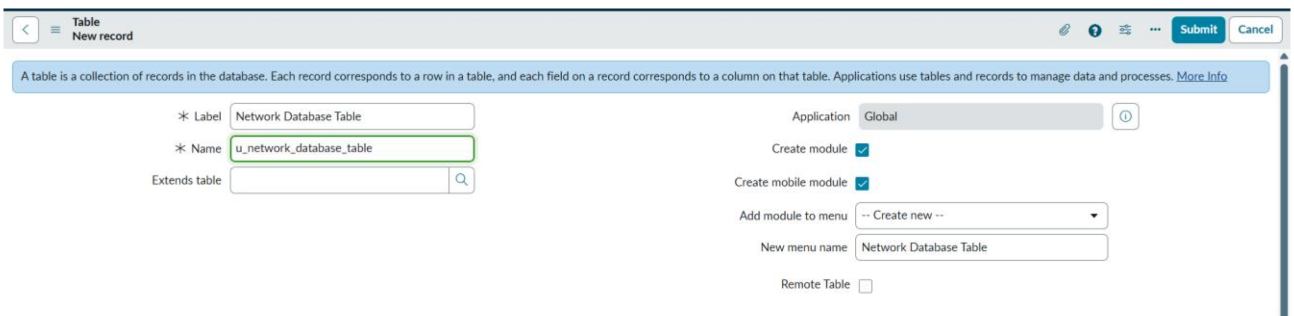


Fig. 9: Network Database Table creation

After the table is created, the Tables list can be used to locate and manage the configured table. This page also demonstrates the ServiceNow table-management environment used during the project.

Purpose of the Table

- Provides a structured destination for request information.
- Allows Flow Designer to create a record from the catalog submission.
- Provides fields that can be updated after approval and notification processing.
- Supports related-list configuration for approval information.

5.2. Fields

- Click All.
- Go to System Definition → Tables.

• Search for Network Database Table.

• Open the table.

• Open the Columns / Fields related list.

• Click New.

• Select the field Type.

• Enter the Column label / Name.

• Configure Reference, Max length, Default value, Display, or other required properties.

• Click Submit.

• Repeat for all required fields.

• Verify that all fields are listed in the table.

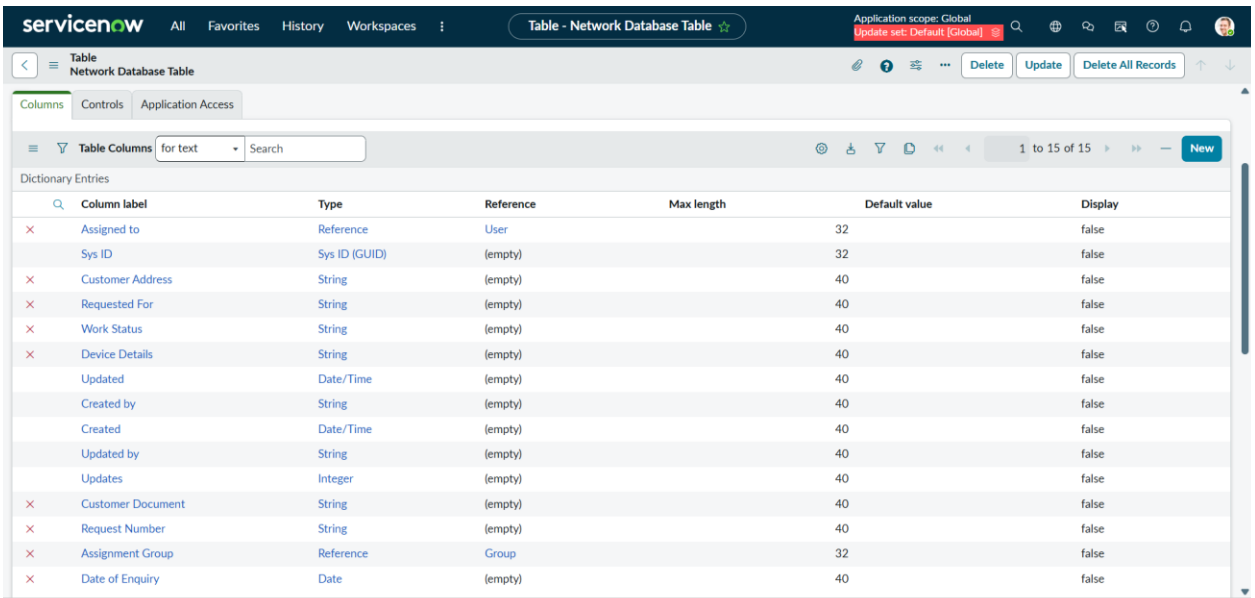


Fig. 11: Network Database Table – Fields / Columns

Field	Description
Assigned to	Reference field used to identify the assigned user.
Customer Address	Stores the customer's address information.
Requested For	Stores the requested-for information.
Work Status	Stores the current work status of the request.
Device Details	Stores device-related information.
Customer Document	Stores customer-document information.
Request Number	Stores the identifier used to track the request.
Assignment Group	Reference field used to identify the responsible group.
Date of Enquiry	Stores the enquiry date.

Field Design Considerations

- Use Reference fields when a field should point to an existing ServiceNow record.

- Use text/string fields for descriptive request information.
- Use date fields for dates that need consistent date-based tracking.
- Use meaningful field labels so the table is easy to maintain and understand.

6. Creation of Relationship

- Click All in the Application Navigator.
- Go to System Definition → Relationships.
- Click New.
- Enter Name as Approval Request.
- Set Applies to table as Network Database Table.
- Set Queries from table to the approval table used in the project.
- Keep the relationship Active.
- Configure the query/script as required.
- Click Submit / Update.

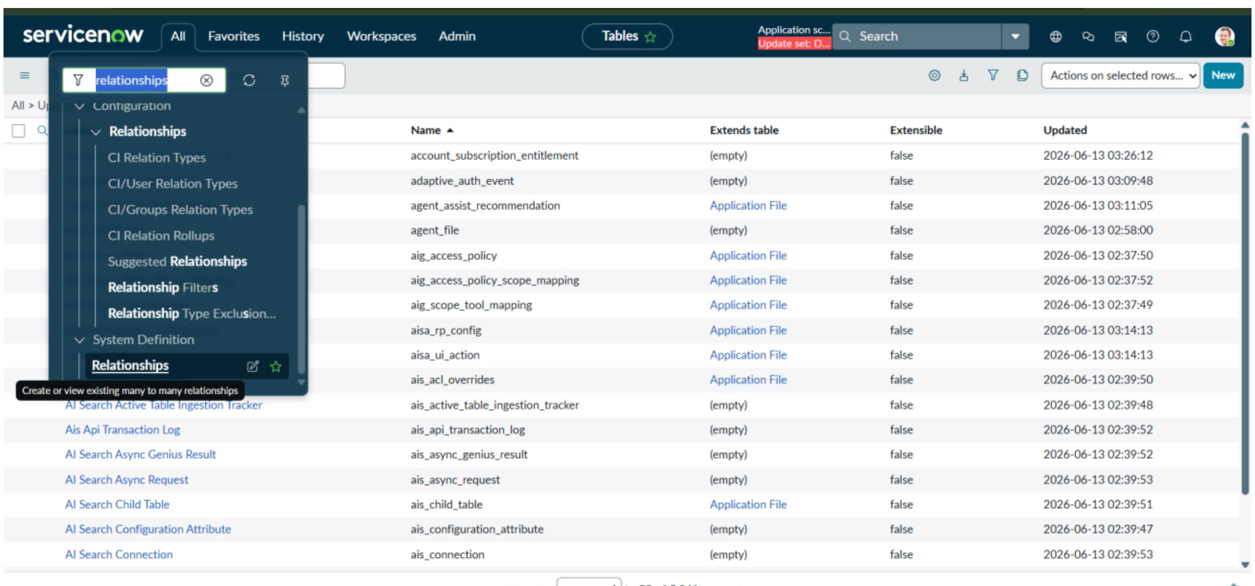


Fig. 12: ServiceNow Relationships search page

6.1. Creation of Related List

Navigate to System Definition > Relationships.

Click New to create a new relationship.

Enter the relationship name as Approval Request.

Set the Applies to table to the Network Database Table.

Set the Queries from table to Approval [sysapproval_approver].

Keep the relationship active.

Save the relationship.

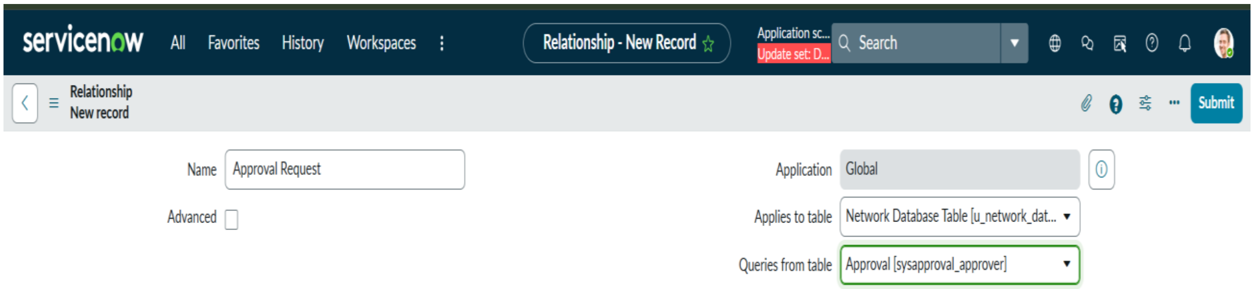


Fig. 13: Approval Request relationship creation

6.2. Relationship Code / Query Script

- Open the Approval Request relationship.
- Locate the Query with / Query Script section.
- Enter the relationship query shown in the screenshot.
- Verify that the query uses the current Network Database Table record.
- Check the script for errors.
- Click Update.
- Verify the approval records through the related list.

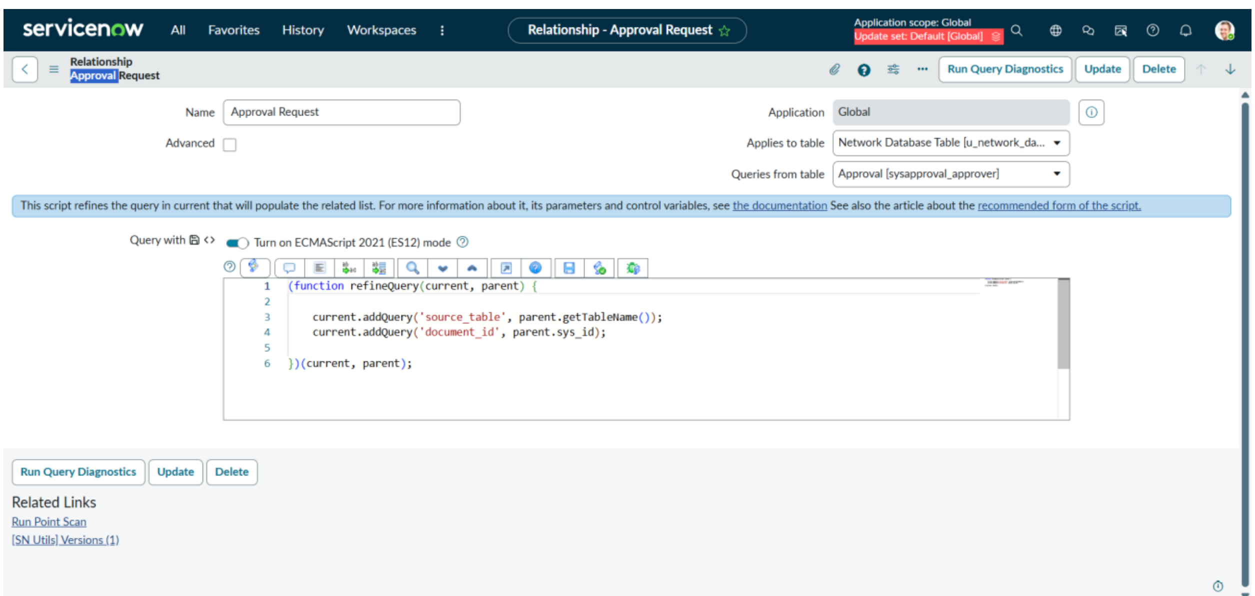


Fig. 13: Approval Request relationship query script / code

9.3 Result

The relationship provides a clear connection between the network request record and its approval information. This improves traceability because approval details can be reviewed in the context of the corresponding request.

7. Creation of Flow

- Click All.

- Go to Process Automation → Flow Designer.
- Open the Network Request flow.
- If it does not exist, click New.
- Enter the flow name and required details.
- Configure the trigger.
- Add the required actions in sequence.
- Save the flow.

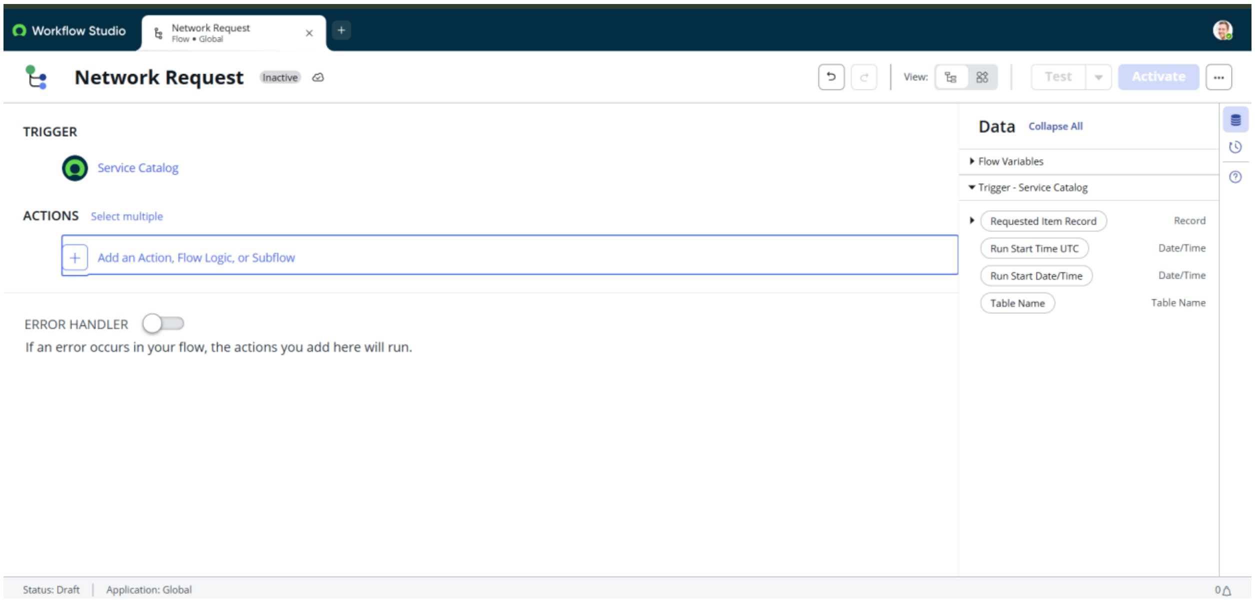


Fig. 14: Network Request Flow Designer

7.2. Get Catalog Variables Creation

- Open the Network Request flow.
- Click Add Trigger.
- Select Service Catalog.
- Configure the Network Request / Requested Item condition.
- Review the trigger.
- Click Save.

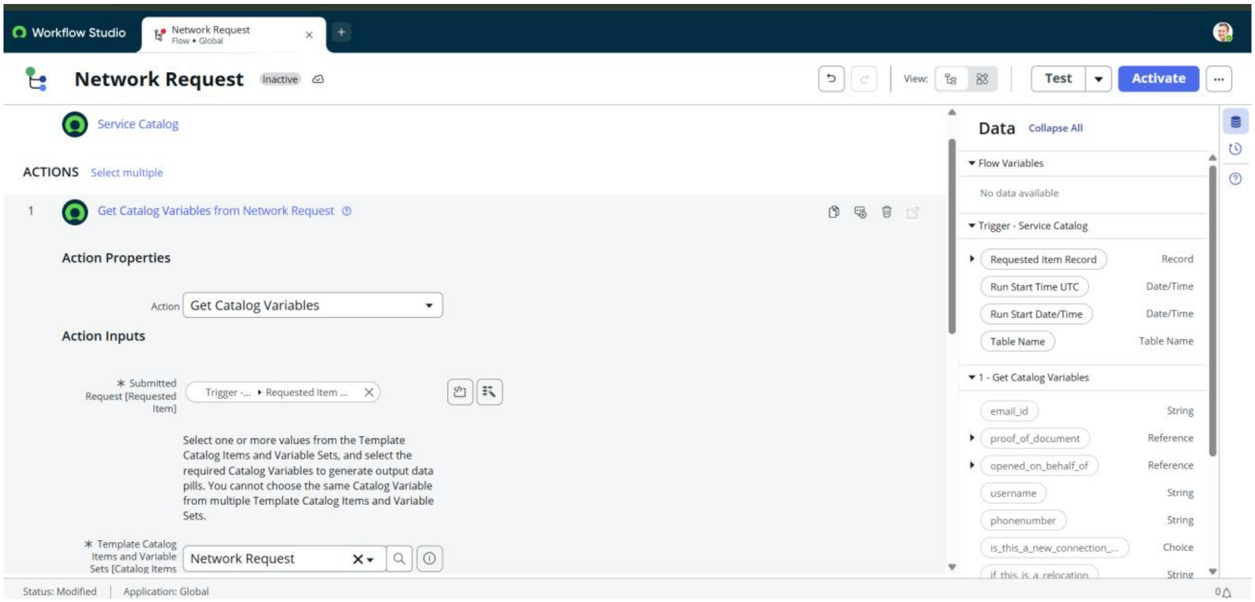


Fig. 15: Service Catalog trigger configuration

- Click Add Action.
- Search for Get Catalog Variables.
- Select the action.
- Provide Requested Item from the trigger.
- Select the required catalog variables.
- Click Save.

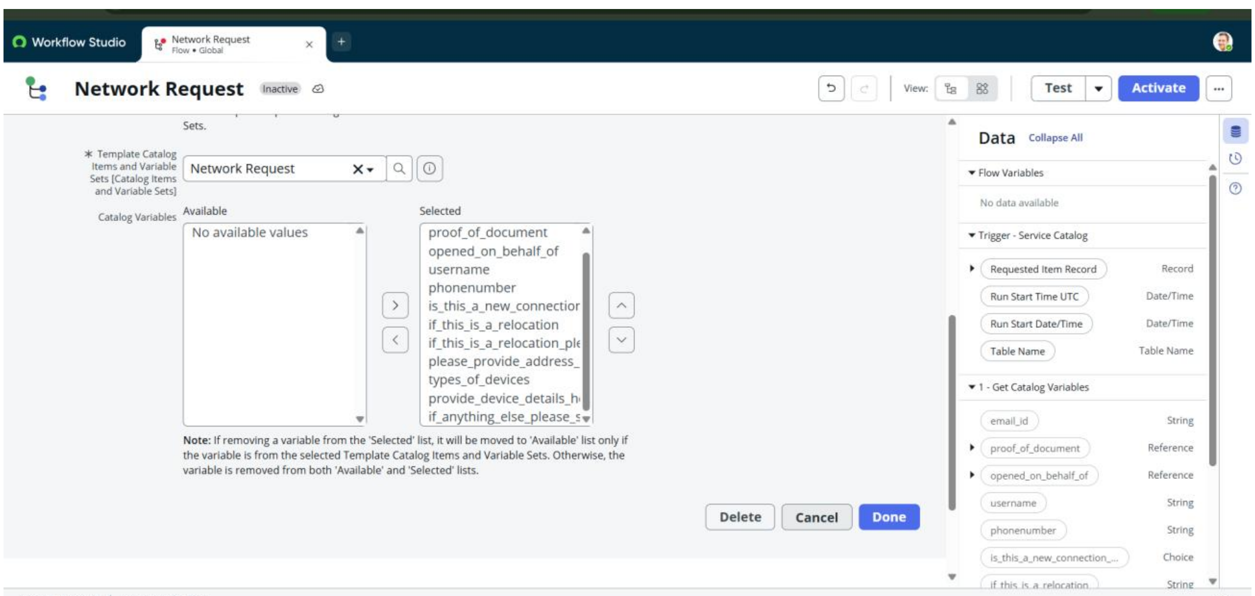


Fig. 16: Get Catalog Variables action

7.3. Create Record

- Click Add Action.
- Select Create Record.

• Select Network Database Table.
• Map the catalog variable data pills to the correct table fields.
• Review every field mapping.
• Click Save.

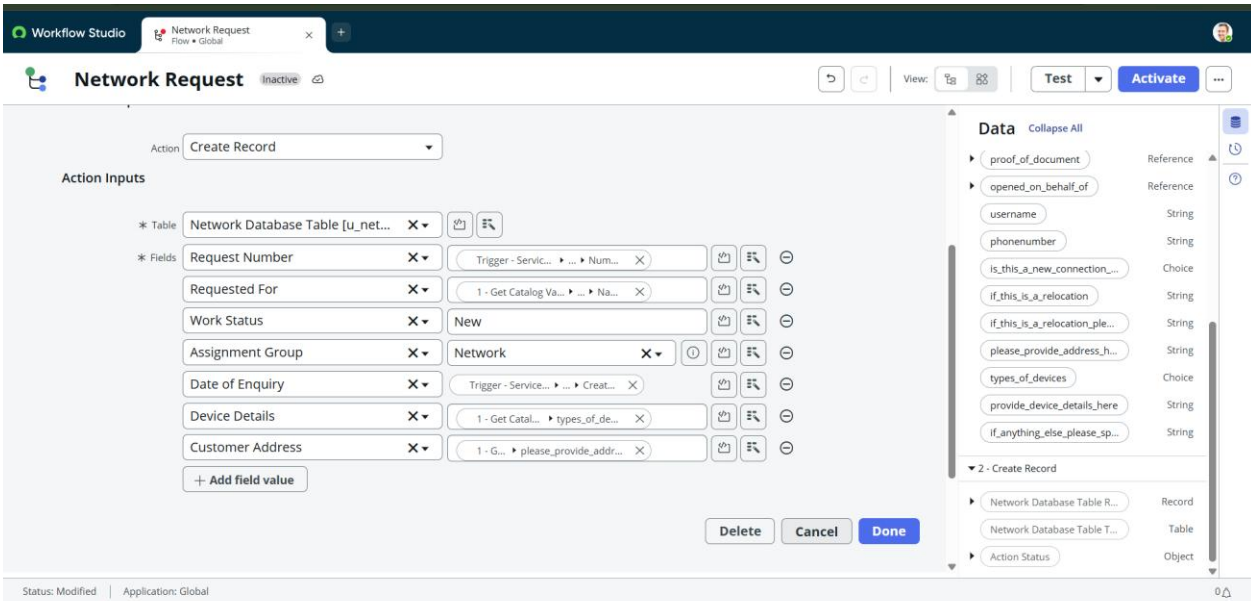


Fig. 17: Create Record action and field mapping

7.4. Ask for Approval Creation

• Add the Approval action after Create Record.
• Configure the approval details.
• Click Add Flow Logic.
• Select If.
• Configure the condition using the approval result.
• Save the condition.

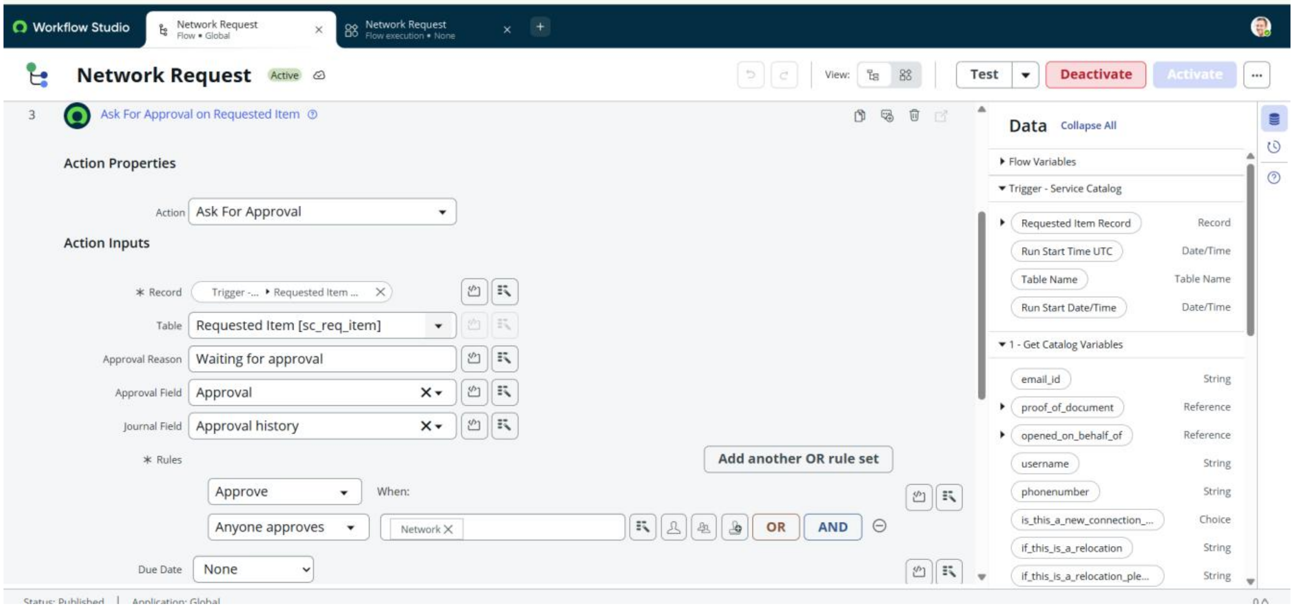


Fig. 18: Approval and conditional processing

7.5. Send Email

- Open the required approval path.
- Click Add Action.
- Select Send Email.
- Configure the recipient.
- Enter the email subject.
- Enter the email message.
- Insert request information using data pills.
- Click Save.

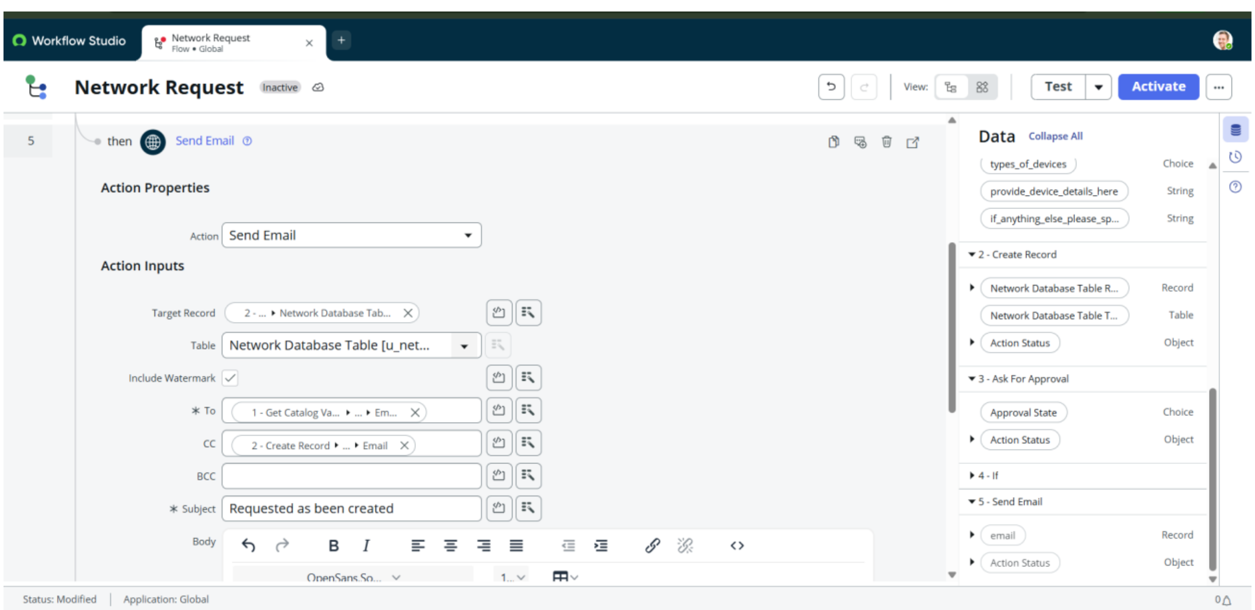


Fig. 19: Send Email action

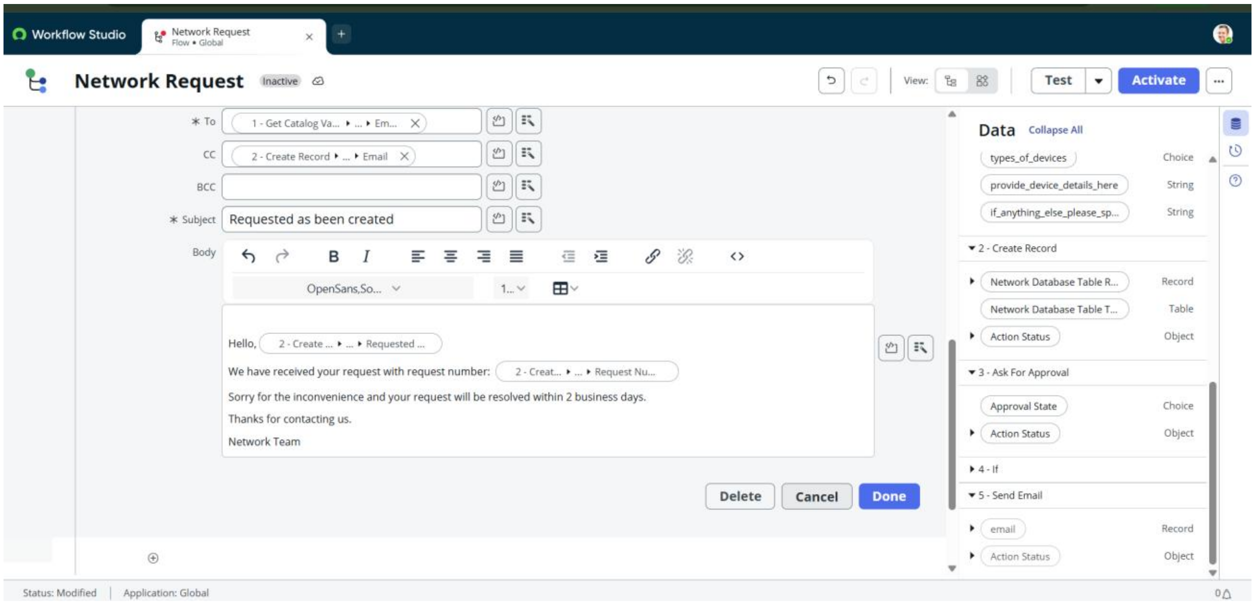


Fig. 20: Email recipient and request-data mapping

7.6. Update Record

- Click Add Action.
- Select Update Record.
- Select the Network Database Table record created earlier.
- Map the fields that must be updated.
- Review the values.
- Click Save.

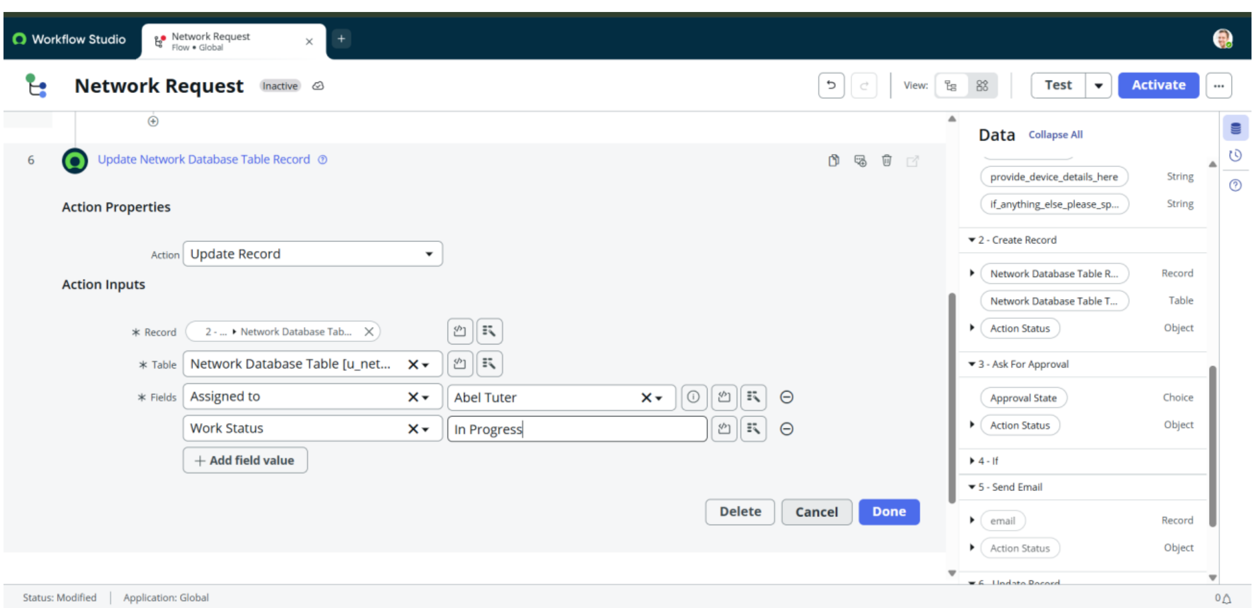


Fig. 21: Update Record action

8. Complete Flow

- Verify Service Catalog Trigger.

- Verify Get Catalog Variables.
- Verify Create Record.
- Verify Approval / Condition.
- Verify Send Email.
- Verify Update Record.
- Check every data-pill mapping.
- Click Save.
- Activate the flow after final verification.

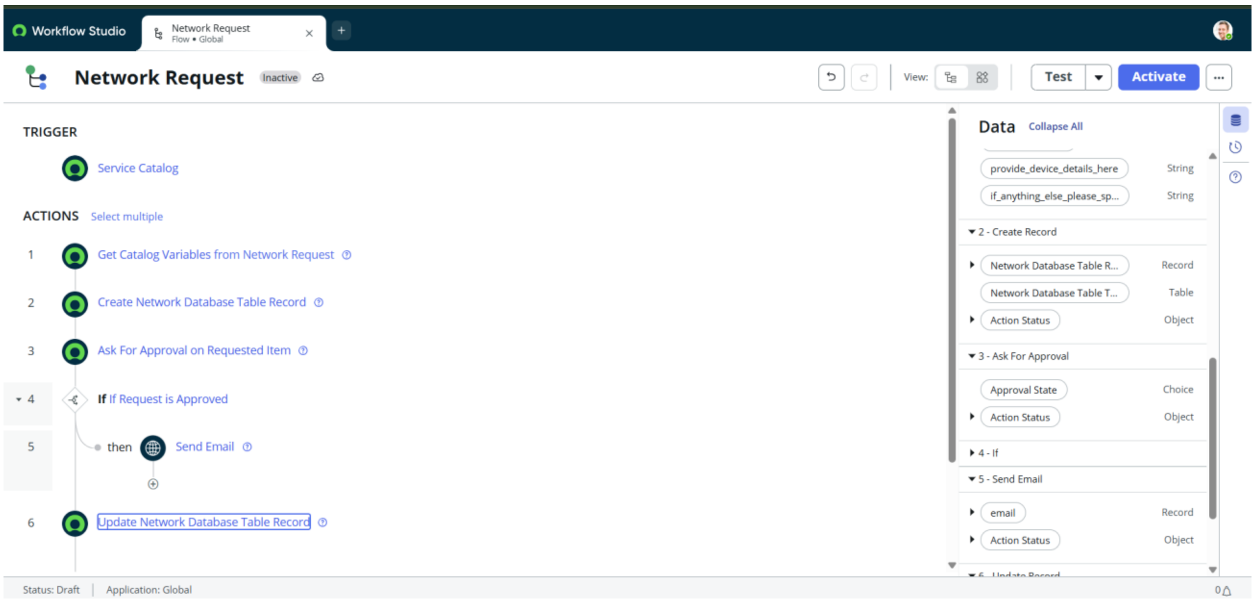


Fig. 22: Complete Network Request flow

9. Catalog Form Output

The catalog form represents the user-facing request interface and shows the fields that are completed during submission.

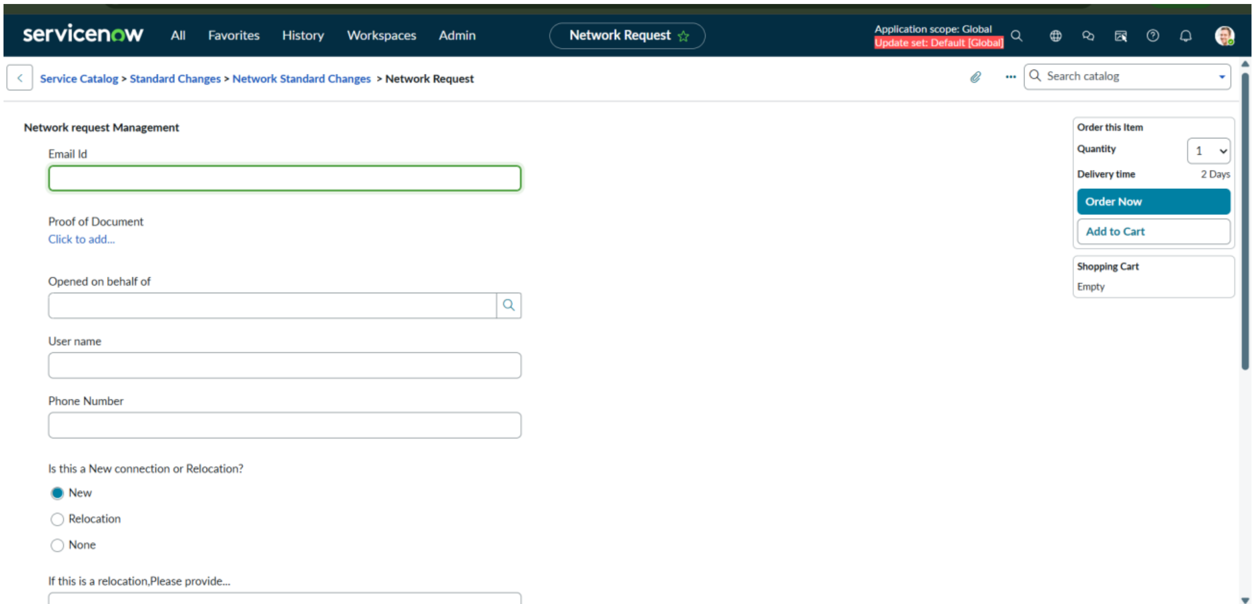


Fig. 26: Network Request catalog form

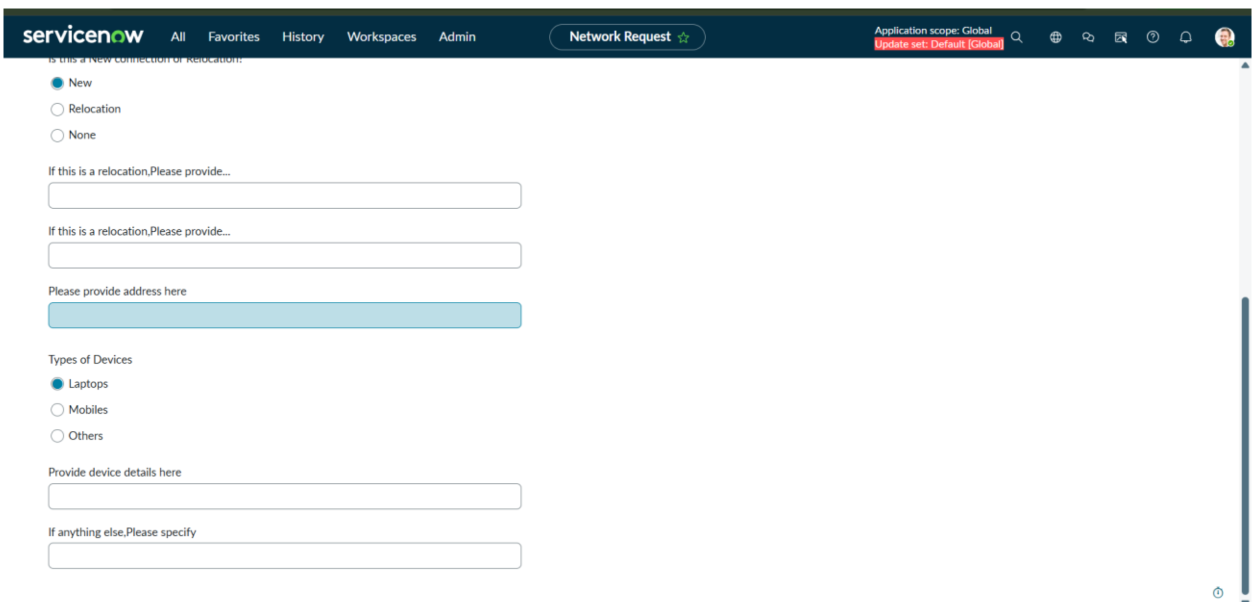


Fig. 27: Network Request completed details

10. Testing the Flow

- Go to All → Process Automation → Flow Designer.
- Open the Network Request flow.
- Click Test.
- Select the required Requested Item test record.
- Start the test.
- Wait for the flow execution to complete.
- Open the execution details.

• Check each action for successful completion.

Test Execution

15. Open the completed Network Request flow.
16. Use the Test option to execute the flow with a suitable request context.
17. Allow the configured actions to run in sequence.
18. Review the execution details for each action.
19. Confirm that no action reports an unexpected failure.

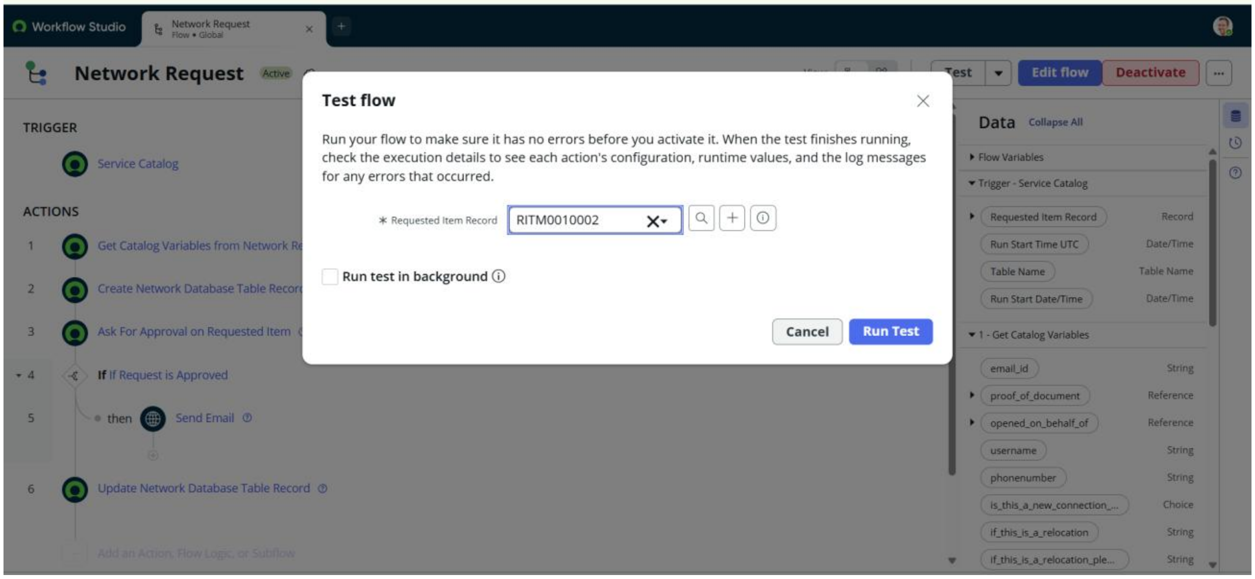


Fig. 23: Test Flow screen

11. Execution Result

The execution result provides evidence that the flow has been run and allows the developer to inspect the processing path and action outcomes.

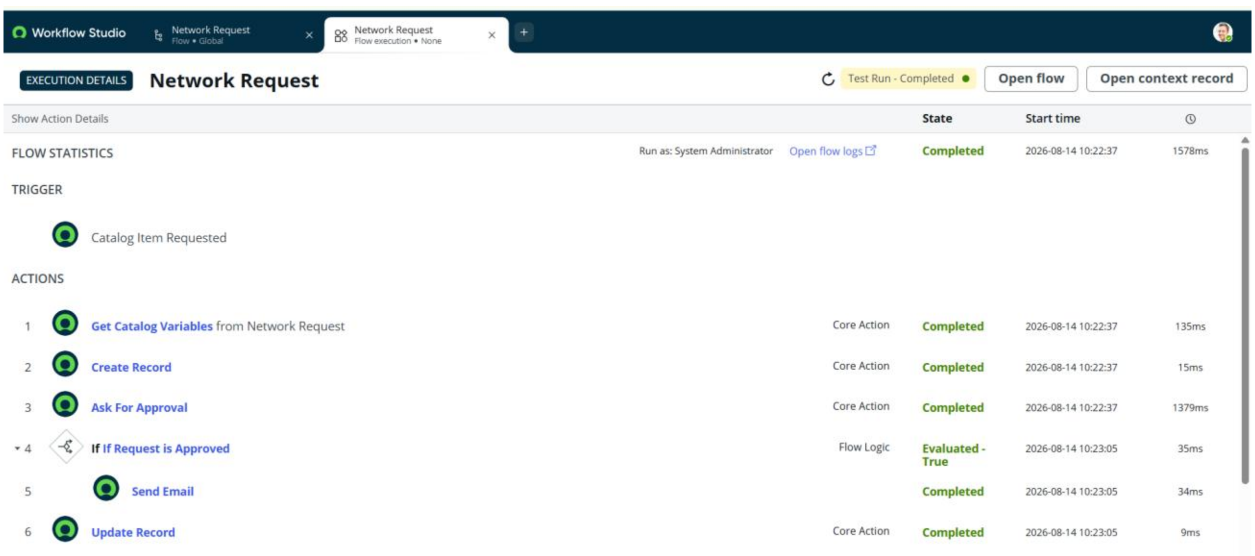
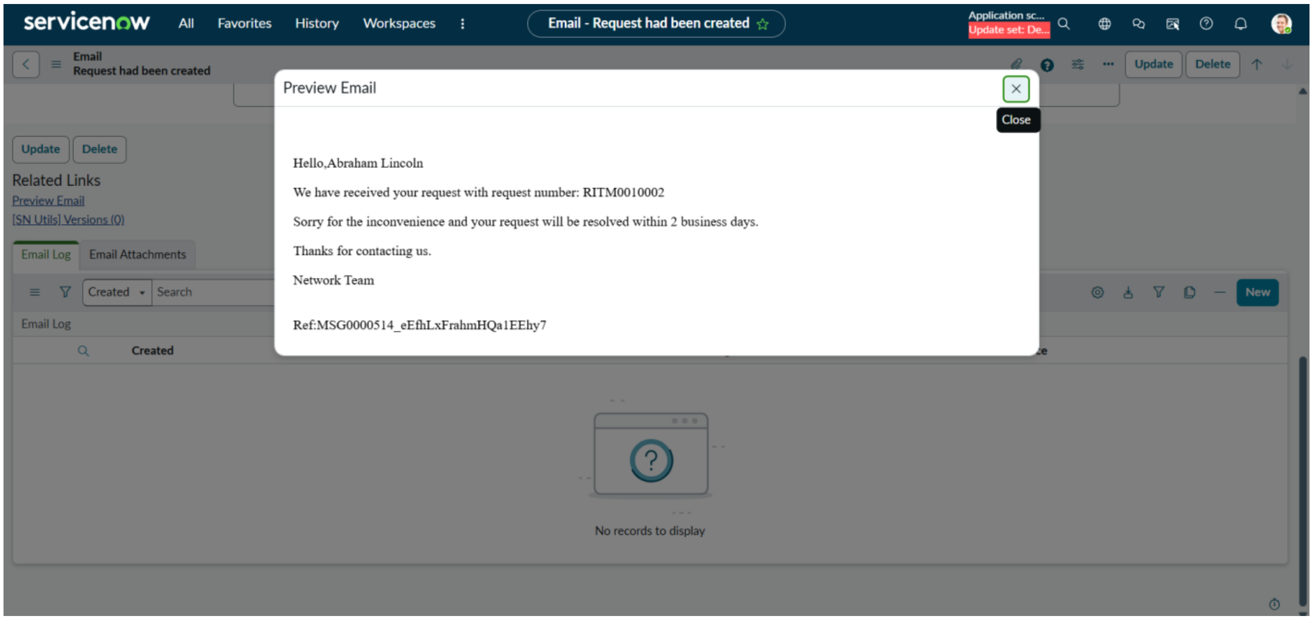


Fig. 24: Flow execution details – Test Run Completed

12. Notification Output



CONCLUSION :

The Automated Network Request Management in ServiceNow project demonstrates how a manual network-request process can be transformed into a structured, self-service, and automated workflow. The implementation starts with the ServiceNow Start Building environment and progresses through catalog configuration, variable design, reusable requester information, dynamic form behavior, backend table configuration, approval relationships, and Flow Designer automation.

- Users receive a structured Network Request catalog form.
- Ten request-specific variables capture the required network information.
- Requester information is organized through a reusable Variable Set containing five variables.
- Two Catalog UI Policies provide dynamic form behavior.
- Network Database Table fields provide a dedicated backend structure.
- Flow Designer automates data retrieval, record creation, approval processing, email notification, and record update.
- Testing and output screenshots provide evidence of the implemented process.

The solution improves consistency, reduces manual handling, supports traceability, and provides a repeatable process for managing network requests. By combining Service Catalog and Flow Designer capabilities, the project provides a clear foundation for scalable and maintainable network service-request management.